

The Kalman Filter: Prior, Likelihood, Posterior

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1 Product of Two Gaussians

Let $p_1(x) \propto \exp(-\frac{(x-\mu_1)^2}{2\sigma_1^2})$ and $p_2(x) \propto \exp(-\frac{(x-\mu_2)^2}{2\sigma_2^2})$ be two Gaussian densities. Their product (up to a normalizing constant) is

$$p(x) \propto p_1(x) p_2(x) = \exp\left(-\frac{(x-\mu_1)^2}{2\sigma_1^2} - \frac{(x-\mu_2)^2}{2\sigma_2^2}\right).$$

Define precisions $\tau_1 = 1/\sigma_1^2$ and $\tau_2 = 1/\sigma_2^2$. Expanding the exponent and completing the square in x gives a quadratic in x with

Precision of the product. $\tau = \tau_1 + \tau_2$ (precisions add), so the posterior variance is

$$\sigma^2 = \frac{1}{\tau} = \frac{1}{\tau_1 + \tau_2} = \frac{1}{1/\sigma_1^2 + 1/\sigma_2^2}.$$

Mean of the product. The precision-weighted average

$$\mu = \frac{\tau_1\mu_1 + \tau_2\mu_2}{\tau_1 + \tau_2} = \frac{\frac{\mu_1}{\sigma_1^2} + \frac{\mu_2}{\sigma_2^2}}{\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}}.$$

So: the product of two Gaussians is Gaussian; the mean is the precision-weighted average of the two means, and the precisions add.

2 Setup: State and Observation Equations

We use the same setup as in the slides:

- State: $\theta_t = \theta_{t-1} + w_t$, $w_t \sim N(0, W)$.
- Observation: $Y_t = \theta_t + v_t$, $v_t \sim N(0, V)$.
- Initial belief: $\theta_0 \sim N(m_0, C_0)$ (prior at time 0).

At each time t , we have a prior $p(\theta_t \mid Y_{1:t-1})$, a likelihood $p(Y_t \mid \theta_t)$, and we want the posterior $p(\theta_t \mid Y_{1:t}) \propto \text{prior} \times \text{likelihood}$. Below we do this explicitly for $t = 1$ and $t = 2$.

3 Step 1: From $t = 0$ to $t = 1$

3.1 Prior for θ_1

Before seeing Y_1 , we have:

- Belief about θ_0 : $N(m_0, C_0)$.
- State equation: $\theta_1 = \theta_0 + w_1$, $w_1 \sim N(0, W)$ independent of θ_0 .

So θ_1 has mean $a_1 = m_0$ and variance $R_1 = C_0 + W$ (variances add for independent sum). The prior for θ_1 is not conditional on any observations (we haven't seen any data yet). Hence

$$\theta_1 \sim N(a_1, R_1), \quad a_1 = m_0, \quad R_1 = C_0 + W.$$

3.2 Likelihood

The observation equation gives $Y_1 = \theta_1 + v_1$, $v_1 \sim N(0, V)$, so

$$Y_1 \mid \theta_1 \sim N(\theta_1, V).$$

3.3 Posterior for θ_1

By Bayes, the posterior is proportional to (prior for θ_1) $\times$ (likelihood of Y_1). To apply the product-of-Gaussians result, we need both factors written as Gaussians in θ_1 :

- Prior: $\theta_1 \sim N(a_1, R_1)$ (mean a_1 , variance R_1).
- Likelihood as a function of θ_1 : The likelihood is $p(Y_1 \mid \theta_1)$, which as a density in Y_1 is $N(\theta_1, V)$. Once we observe Y_1 , we fix that number and view the likelihood as a function of the unknown θ_1 . So $L(\theta_1) \propto \exp\left(-\frac{(Y_1 - \theta_1)^2}{2V}\right) = \exp\left(-\frac{(\theta_1 - Y_1)^2}{2V}\right)$. That is a Gaussian in θ_1 with mean Y_1 and variance V . So the likelihood term (as a function of θ_1) is centered at the observation Y_1 because the exponent is $(\theta_1 - Y_1)^2$: the most plausible value of θ_1 given the single observation Y_1 is $\theta_1 = Y_1$, and uncertainty around that is V .

So we are combining $N(a_1, R_1)$ with $N(Y_1, V)$ (the latter centered at the observation). Using the product-of-Gaussians result with $\mu_1 = a_1$, $\sigma_1^2 = R_1$, $\mu_2 = Y_1$, $\sigma_2^2 = V$:

$$\frac{1}{C_1} = \frac{1}{R_1} + \frac{1}{V}, \quad \text{so } C_1 = \frac{R_1 V}{R_1 + V} = (1 - K_1)R_1, \quad K_1 = \frac{R_1}{R_1 + V}.$$

$$m_1 = \frac{a_1/R_1 + Y_1/V}{1/R_1 + 1/V} = a_1 + K_1(Y_1 - a_1).$$

So

$$\theta_1 \mid Y_1 \sim N(m_1, C_1), \quad m_1 = a_1 + K_1(Y_1 - a_1), \quad C_1 = (1 - K_1)R_1, \quad K_1 = \frac{R_1}{R_1 + V}.$$

4 Step 2: From $t = 1$ to $t = 2$

4.1 Prior for θ_2

We now have the posterior at $t = 1$: $\theta_1 \mid Y_1 \sim N(m_1, C_1)$. The state equation is $\theta_2 = \theta_1 + w_2$, $w_2 \sim N(0, W)$. So θ_2 has mean $a_2 = m_1$ and variance $R_2 = C_1 + W$:

$$\theta_2 \mid Y_1 \sim N(a_2, R_2), \quad a_2 = m_1, \quad R_2 = C_1 + W.$$

4.2 Likelihood

$$Y_2 \mid \theta_2 \sim N(\theta_2, V).$$

4.3 Posterior for θ_2

Same as before: the likelihood as a function of θ_2 is $L(\theta_2) \propto \exp(-\frac{(\theta_2 - Y_2)^2}{2V})$, i.e., a Gaussian in θ_2 centered at the observation Y_2 with variance V .

Combine prior $N(a_2, R_2)$ with that. So

$$\frac{1}{C_2} = \frac{1}{R_2} + \frac{1}{V}, \quad C_2 = (1 - K_2)R_2, \quad K_2 = \frac{R_2}{R_2 + V}.$$

$$m_2 = a_2 + K_2(Y_2 - a_2).$$

$$\theta_2 \mid Y_1, Y_2 \sim N(m_2, C_2), \quad m_2 = a_2 + K_2(Y_2 - a_2), \quad C_2 = (1 - K_2)R_2, \quad K_2 = \frac{R_2}{R_2 + V}.$$

5 Summary

- **Product of two Gaussians:** Gaussian with precision-weighted mean and precisions add ($1/\sigma^2 = 1/\sigma_1^2 + 1/\sigma_2^2$).
- $t = 0 \rightarrow 1$: Prior $N(m_0, C_0)$ and state eqn $\Rightarrow$ prior for θ_1 : $N(a_1, R_1)$ with $a_1 = m_0$, $R_1 = C_0 + W$. Likelihood $Y_1 \mid \theta_1 \sim N(\theta_1, V)$. Posterior $\theta_1 \mid Y_1 \sim N(m_1, C_1)$ with $m_1 = a_1 + K_1(Y_1 - a_1)$, $C_1 = (1 - K_1)R_1$, $K_1 = R_1/(R_1 + V)$.
- $t = 1 \rightarrow 2$: Prior for θ_2 : $N(a_2, R_2)$ with $a_2 = m_1$, $R_2 = C_1 + W$. Likelihood $Y_2 \mid \theta_2 \sim N(\theta_2, V)$. Posterior $\theta_2 \mid Y_1, Y_2 \sim N(m_2, C_2)$ with $m_2 = a_2 + K_2(Y_2 - a_2)$, $C_2 = (1 - K_2)R_2$, $K_2 = R_2/(R_2 + V)$.

The same pattern repeats for $t = 2 \rightarrow 3$, $t = 3 \rightarrow 4$, etc.: predict (prior mean = last posterior mean, prior variance = last posterior variance + W), then update (posterior = precision-weighted combination of prior and likelihood).